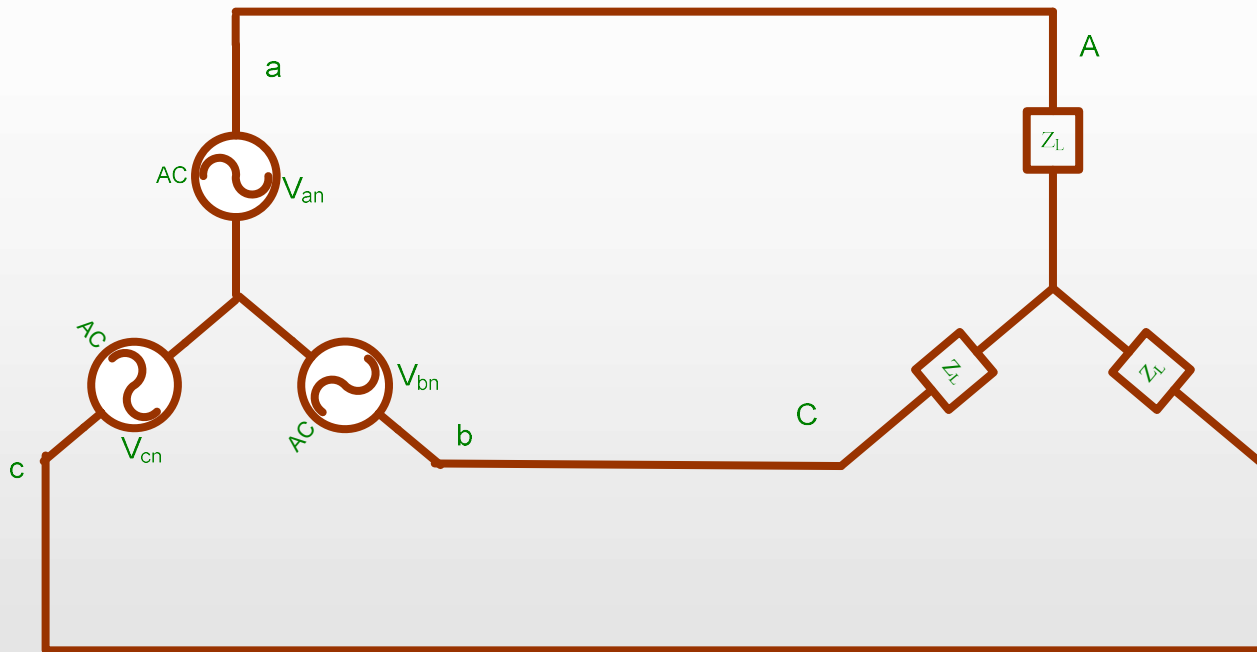


LECTURE –VII
THREE-PHASE SUPPLY SYSTEMS

Wye – Wye System



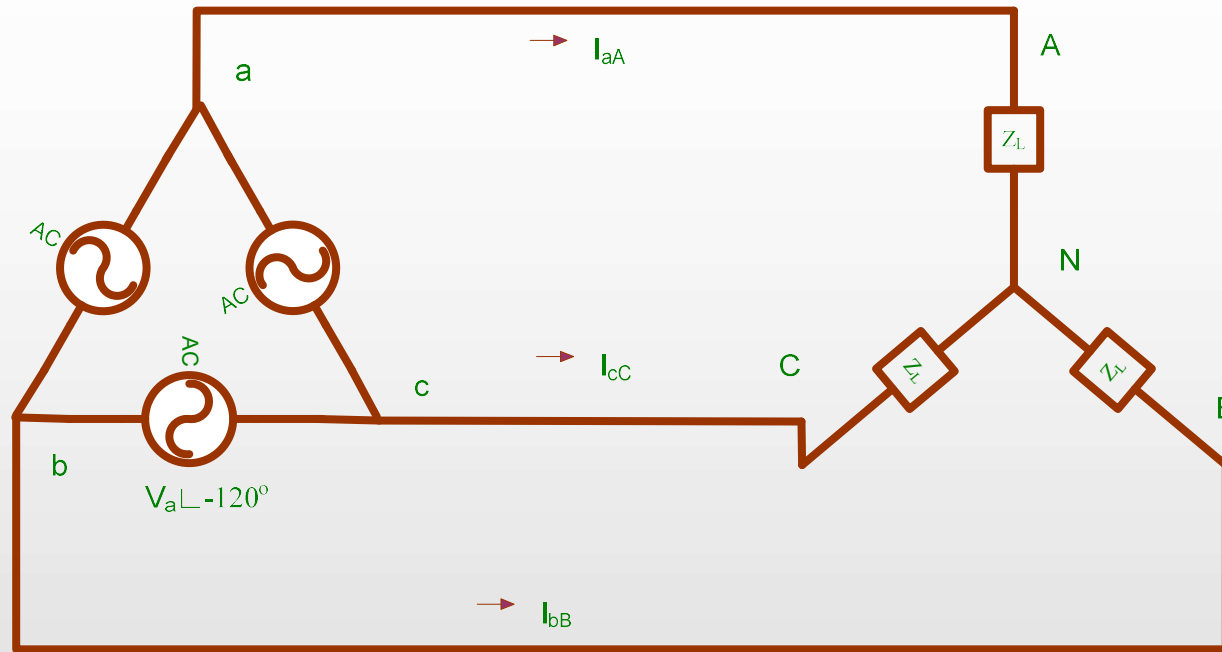
For balanced load impedance, the load phase voltage and line voltage are related as

and $V_{AB} = \sqrt{3} V_{AN}$, $V_{BC} = \sqrt{3} V_{BN}$, $V_{CA} = \sqrt{3} V_{CN}$,

however load phase currents and line currents are related as

$I_{aA} = I_{AN}$, $I_{bB} = I_{BN}$, $I_{cC} = I_{CN}$,

Delta – Wye System



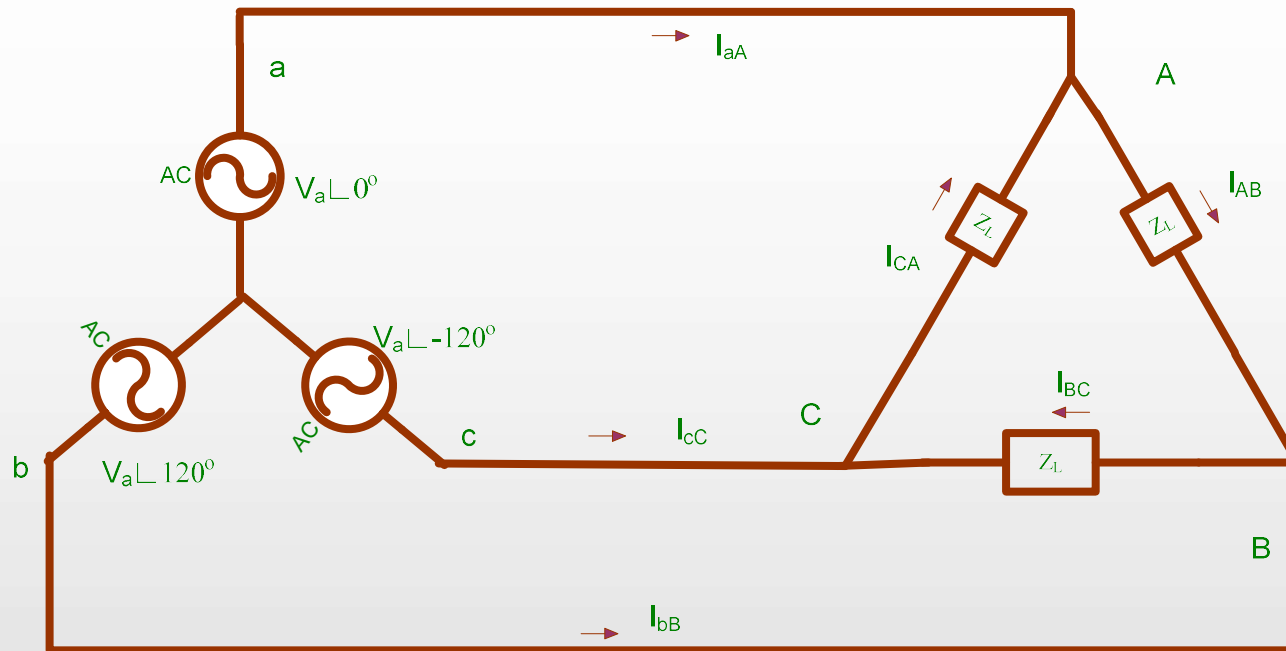
The Load phase voltage and Load line voltage are related for balanced load as ,

$$V_{AB} = \sqrt{3} V_{AN}, V_{BC} = \sqrt{3} V_{BN}, V_{CA} = \sqrt{3} V_{CN},$$

The Load Phase currents and Line Currents are related as,

$$I_{aA} = I_{AN}, I_{bB} = I_{BN}, I_{cC} = I_{CN},$$

Wye-Delta System



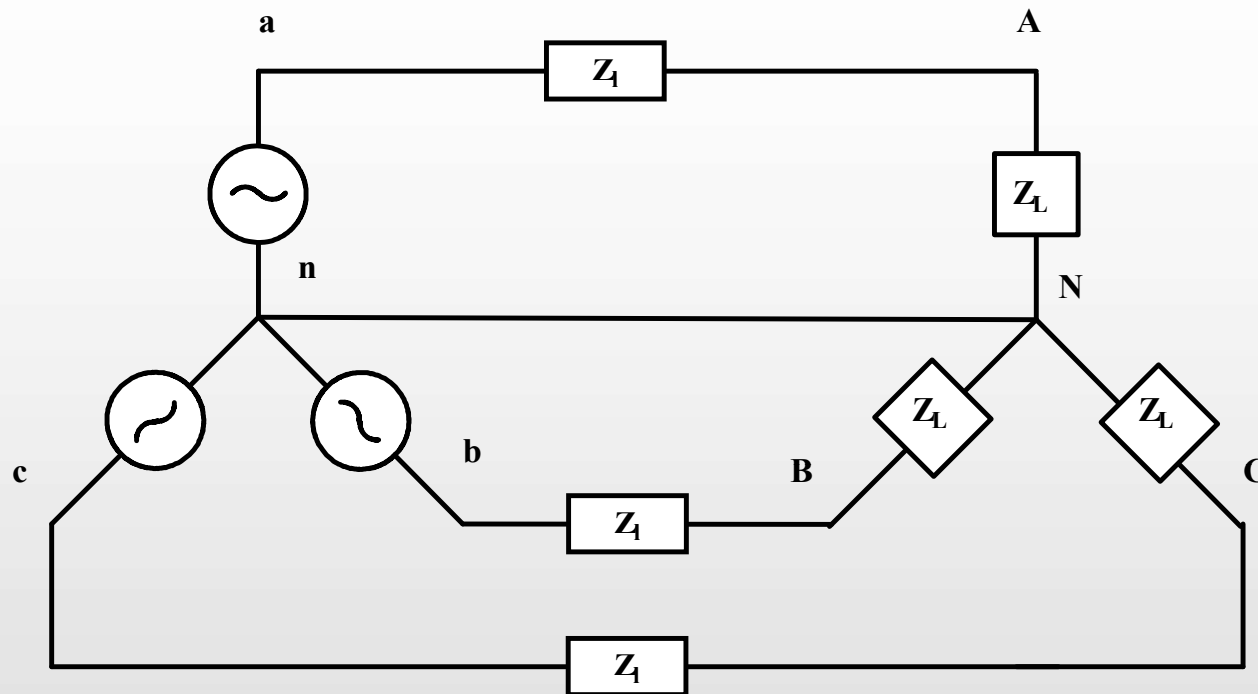
The Load phase voltage and Load line voltage are related as ,

Line voltage = Phase voltage

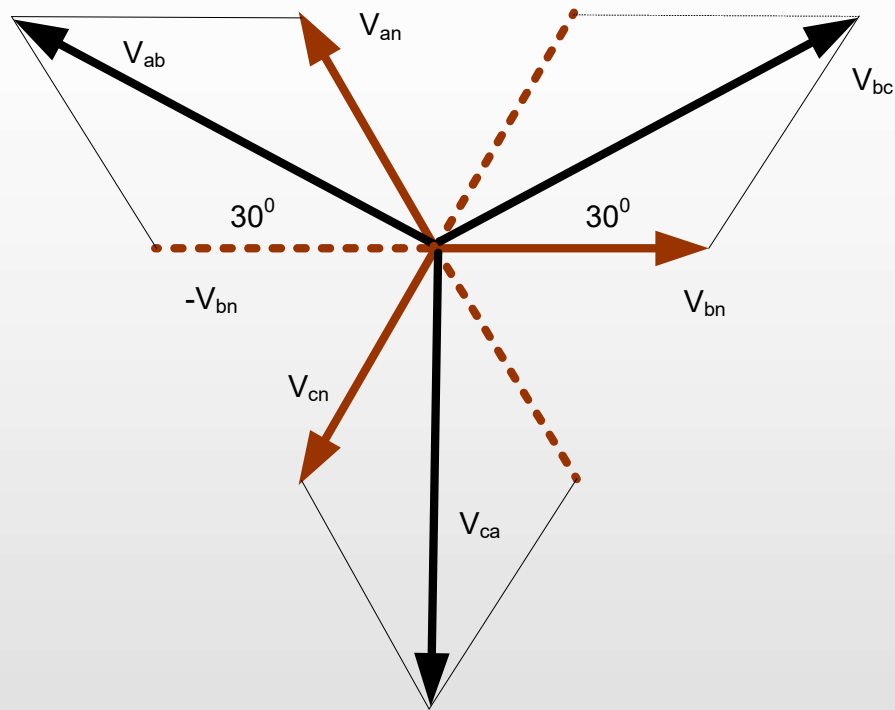
The Load Phase currents and Line Currents are related as,

Line Current = $\sqrt{3}$ * Phase current

Wye – Wye System



Considering the effect of source impedance and applying the KVL
 $V_{an} = V_{AN} + I_A Z_l$; $V_{bn} = V_{BN} + I_B Z_l$, $V_{cn} = V_{CN} + I_C Z_l$ and for balanced
 load impedance, the load phase voltage and line voltage are related
 as $V_{AB} = \sqrt{3} V_{AN}$, $V_{BC} = \sqrt{3} V_{BN}$, $V_{BC} = \sqrt{3} V_{CN}$, however currents are
 related as $I_{AB} = I_{AN}$, $I_{BC} = I_{BN}$, $I_{CA} = I_{CN}$.



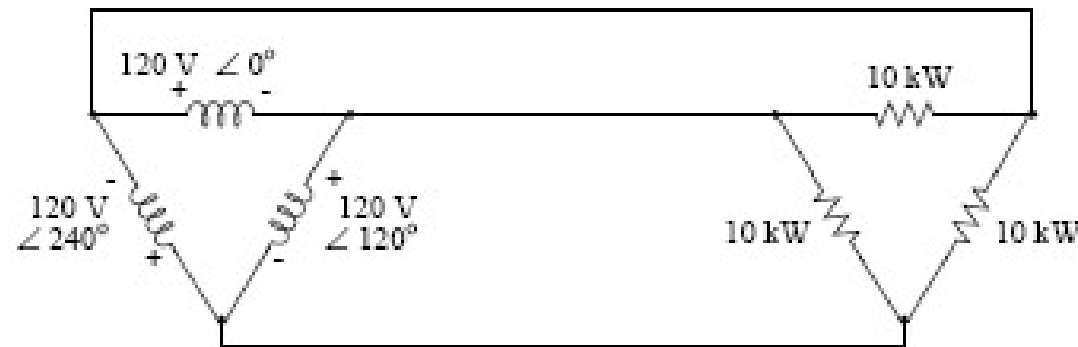
$$\begin{aligned}
 V_{ab} &= V_{an} - V_{bn} \\
 &= |V_p| \angle 0^\circ - |V_p| \angle -120^\circ \\
 &= |V_p| (1 - (\cos 120 - j \sin 120)) \\
 &= |V_p| - |V_p| \left(\frac{1}{2} - j \frac{\sqrt{3}}{2} \right) \\
 &= \sqrt{3} |V_p| \angle 30^\circ
 \end{aligned}$$

$$V_{bc} = \sqrt{3} |V_p| \angle -90^\circ$$

$$V_{ca} = \sqrt{3} |V_p| \angle -210^\circ$$

$$V_L = \sqrt{3} |V_p| = \text{Line Voltage}$$

Example



For D ("delta") balanced system-

$$V_{\text{line}} = V_{\text{phase}}$$

$$I_{\text{line}} = \sqrt{3} I_{\text{phase}}$$

With each load resistance receiving 120 volts from its respective phase winding at the source, the current in each phase of this circuit will be $(10 \times 10^3 / 120) = 83.33$ amps. Now,

$$I_{\text{line}} = \sqrt{3} I_{\text{phase}}$$

$$I_{\text{line}} = \sqrt{3} \times 83.33 = 144.34 \text{ A}$$

Ex. Three impedances of $3+j4 \Omega$ are connected in delta to a 220V, three phase line. Find the line current, the power factor and the complex power

Sol.

Given $Z_P = 3 + j4 = 5 \angle 53.1^\circ$

$$I_P = \frac{V_P}{Z_P} = \frac{220}{5} = 44 A$$

$$I_L = \sqrt{3} * 44 = 76.2 A$$

$$p.f. = \cos \theta = \cos 53.1^\circ = 0.6 \text{ lagging}$$

$$\sin \theta = 0.8$$

$$S = \sqrt{3} V_L I_L = \sqrt{3} \cdot 220 \cdot 76.2 = 29000 VA$$

$$S = 17400 + j23,200 VA$$

Ex. Three impedances of $4-j3 \Omega$ are connected in star to a 208V, three phase line. Find the line current, the power factor and the complex power

Sol. $Z_P = 4 - j3 = 5 \angle -36.9^\circ$

$$V_P = \frac{208}{\sqrt{3}} = 120V$$

$$I_L = I_P = \frac{V_P}{Z_P} = \frac{120}{5} = 24A$$

$$p.f. = \cos \theta = \cos(-36.9)^\circ = 0.8, \text{leading}$$

$$\sin \theta = 0.8$$

$$S = \sqrt{3}V_L I_L = \sqrt{3} \cdot 208 \cdot 24 = 8.65KVA$$

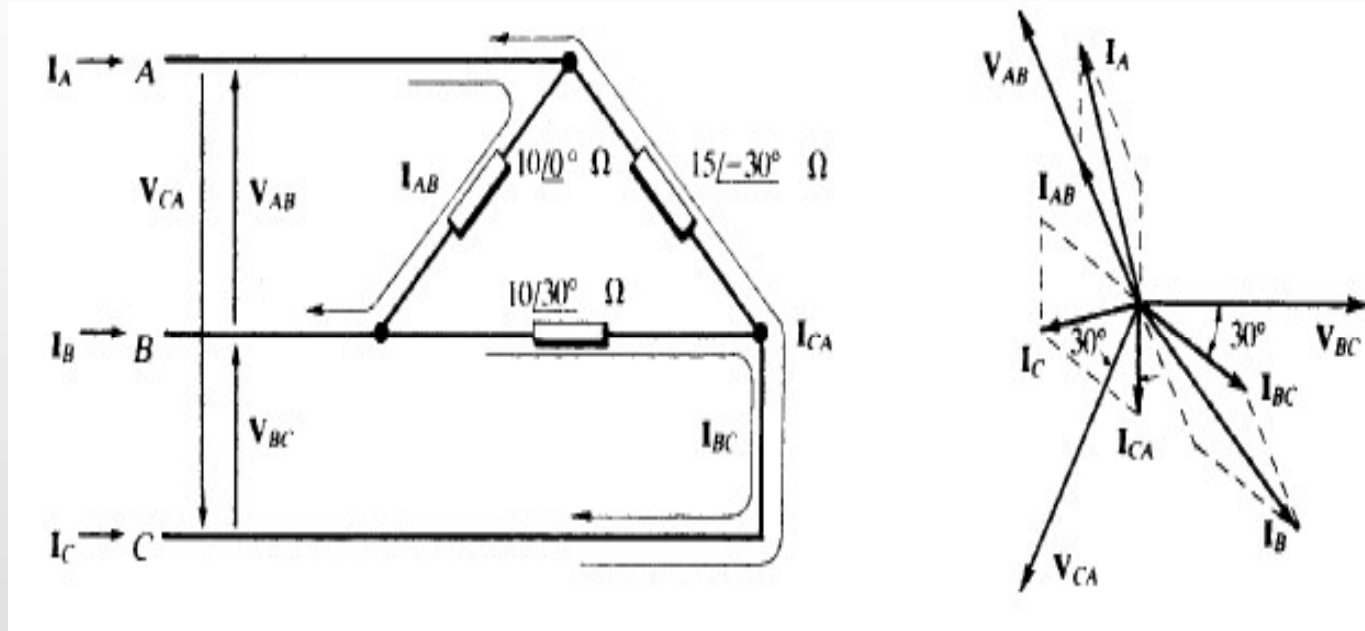
$$P = S \cos(-36.9^\circ) = 6.92kW$$

$$Q = S \sin(-36.9^\circ) = -5.19kVAR$$

$$S = P - jQ = 6.92 - 5.19KVA$$

Unbalanced Loads in Three- Phase Systems

A three-phase, 339.4V, ABC system has a delta-connected load, with $Z_{AB} = 10 \angle 0^\circ$, $Z_{BC} = 10 \angle 30^\circ$, $Z_{CA} = 15 \angle -30^\circ$. Obtain phase and line currents and draw the phasor diagram



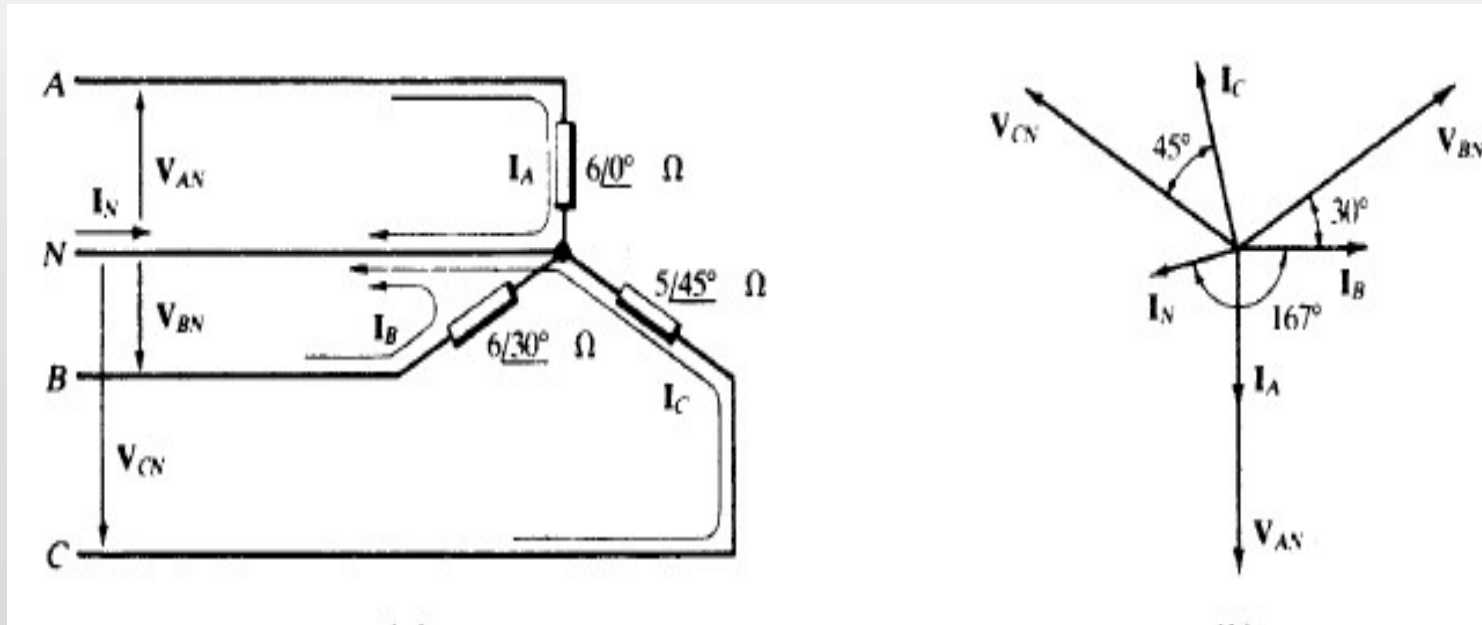
$$I_{AB} = \frac{V_{AB}}{Z_{AB}} = \frac{339.4 \angle 120^\circ}{10 \angle 0^\circ} = 33.94 \angle 120^\circ;$$

$$\text{Similarly } I_{BC} = 33.94 \angle -30^\circ; I_{CA} = 22.63 \angle 270^\circ;$$

$$I_A = I_{AB} + I_{AC} = 33.94 \angle 120^\circ - 22.63 \angle 270^\circ = 54.72 \angle 108.1^\circ$$

$$\text{Also } I_B = 65.56 \angle -45^\circ, \text{ and } I_C = 29.93 \angle -169.1^\circ$$

Ex. A three-phase, four-wire, 150-V, CBA system has a Y-connected load, with $Z_A = 6 \angle 0^\circ$, $Z_B = 6 \angle 30^\circ$, $Z_C = 5 \angle 45^\circ$. Obtain all line currents and draw the phasor diagram.



$$I_A = \frac{V_{AN}}{Z_A} = \frac{86.6 \angle 90^\circ}{6 \angle 0^\circ} = 14.43 \angle 90^\circ$$

$$I_B = \frac{V_{BN}}{Z_B} = \frac{86.6 \angle 30^\circ}{6 \angle 30^\circ} = 14.43 \angle 0^\circ$$

$$I_C = \frac{V_{CN}}{Z_C} = \frac{86.6 \angle 150^\circ}{5 \angle 45^\circ} = 17.32 \angle 105^\circ$$

$$\begin{aligned} \text{so } I_N &= -(14.43 \angle 90^\circ + 14.43 \angle 0^\circ + 17.32 \angle 105^\circ) \\ &= 10.21 \angle -167^\circ \end{aligned}$$

Ex. For the system shown in Fig 1. obtain the line currents and find the displacement neutral voltage, V_{ON} . The supply line voltage is 150 V

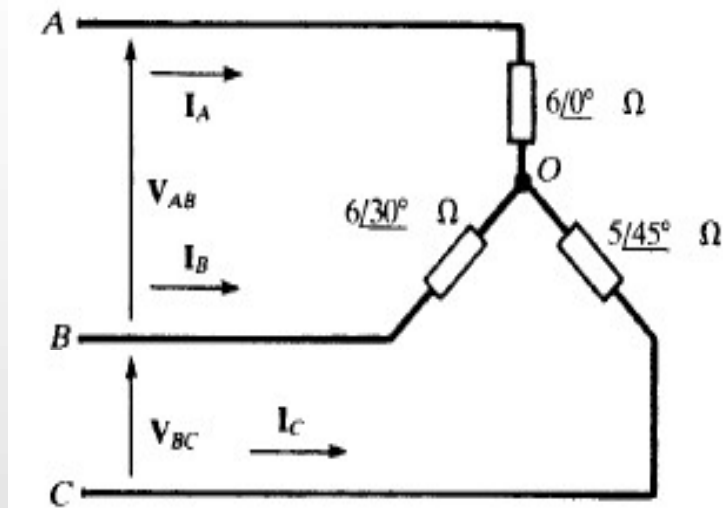


Fig. 1

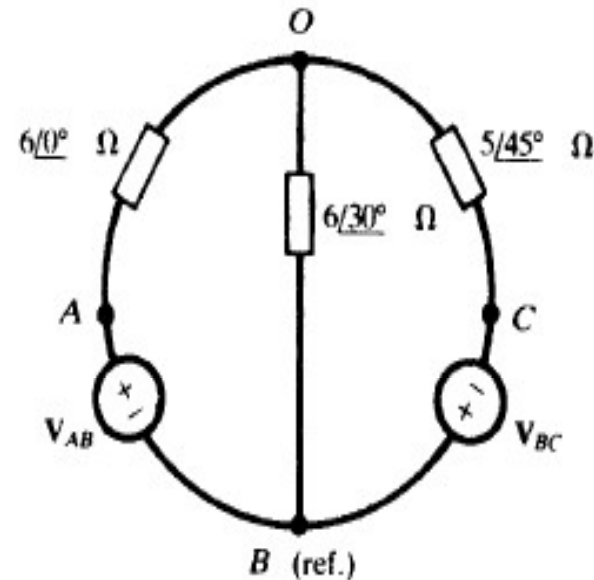


Fig. 2

The circuit is redrawn in Fig. 2 so as to suggest a single node-voltage equation with V_{OB} as the unknown.

$$\frac{V_{OB} - V_{AB}}{Z_A} + \frac{V_{OB}}{Z_B} + \frac{V_{OB} + V_{BC}}{Z_C} = 0$$

$$V_{OB} \left(\frac{1}{6 \angle 0^\circ} + \frac{1}{6 \angle 30^\circ} + \frac{1}{5 \angle 45^\circ} \right) = \frac{150 \angle 240^\circ}{6 \angle 0^\circ} - \frac{150 \angle 0^\circ}{5 \angle 45^\circ}$$

from which $V_{OB} = 66.76 \angle -152.85^\circ V$

$$\text{Then } I_B = \frac{V_{OB}}{Z_B} = 11.13 \angle -2.85^\circ A$$

$$\text{From } V_{OA} + V_{AB} = V_{OB}, \quad V_{OA} = 100.7 \angle -81.05^\circ V$$

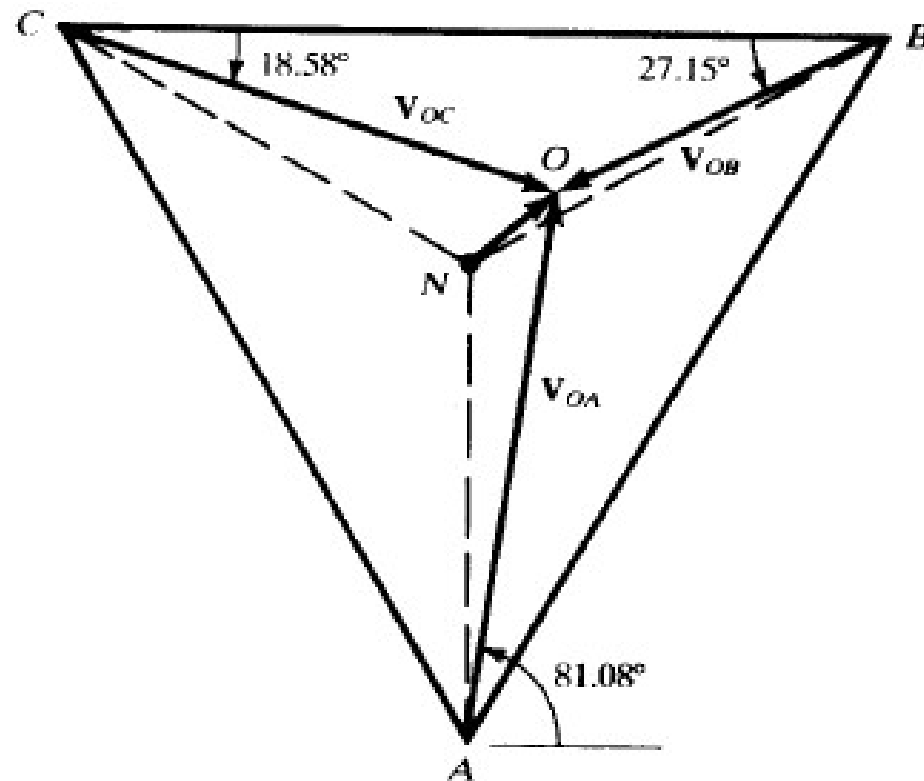
$$I_A = -\frac{V_{OA}}{Z_A} = 16.78 \angle -98.92^\circ A$$

$$\text{Similarly } V_{OC} = V_{OB} - V_{CB}, \quad V_{OA} = 95.58 \angle -18.58^\circ V$$

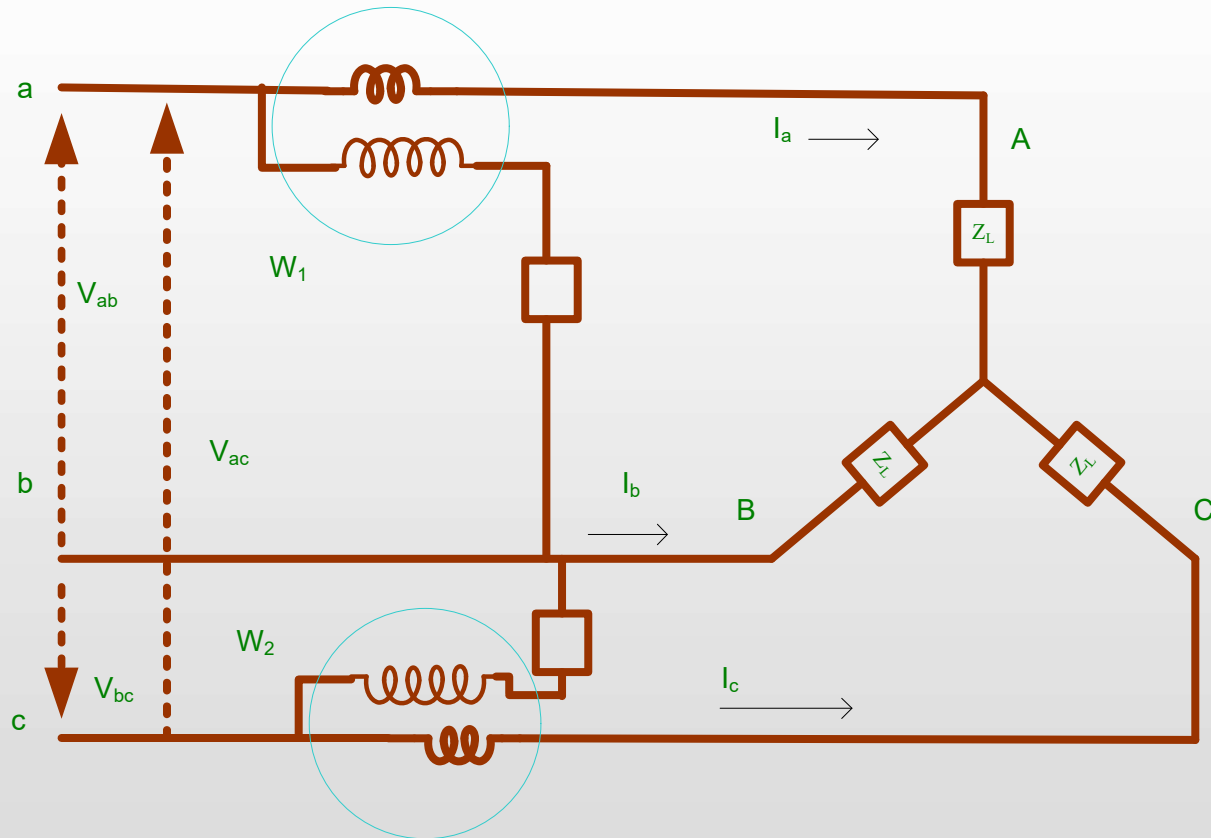
$$I_C = -\frac{V_{OC}}{Z_C} = 19.12 \angle 116^\circ A$$

Point O is displaced from the neutral N by a phasor voltage V_{ON} , given by

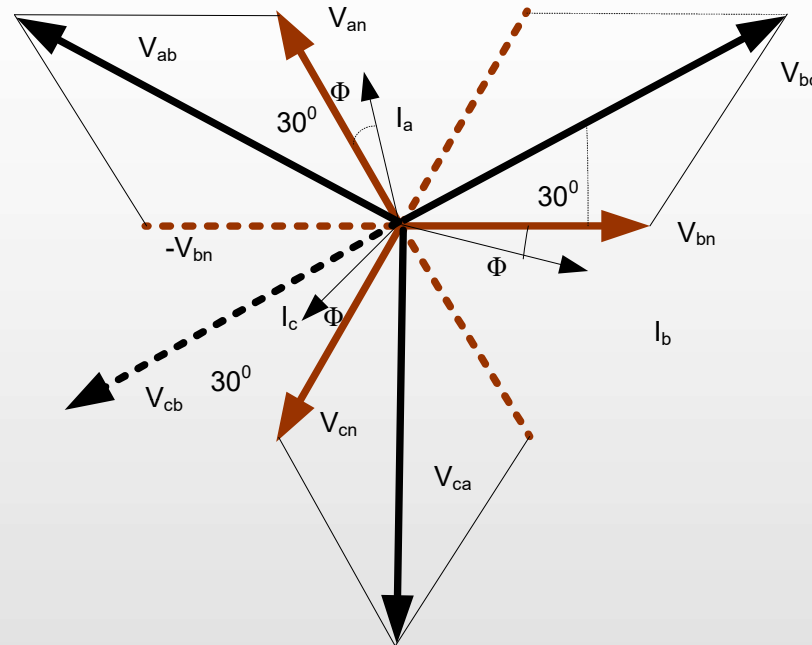
$$V_{ON} = V_{OA} + V_{AN} = 100.7 \angle 81.08^\circ + \frac{150}{\sqrt{3}} \angle 90^\circ = 20.24 \angle 39.53^\circ V$$



Power Measurement by means of two Wattmeter's



Phasor Diagram



Assume sinusoidal voltages and current, so they can be represented by phasor.

The currents being assume to lag the corresponding phase voltages by an angle Φ . Current through current coil of W_1 is I_a and potential difference across voltage circuit of W_1 is V_{ab}

Phase difference of V_{an} and $V_{bn} = V_{ab}$

Phase difference between I_a and $V_{ab} = 30^\circ + \Phi$.

therefore the reading on W_1 , which depends on line current, line voltage and phase angle between line current and line voltage is

$$P_1 = V_{ab} I_a \cos (30^\circ + \Phi)$$

Similarly,

$$P_2 = V_{cb} I_c \cos (30^\circ - \Phi)$$

Since load is balanced

$$\text{So } I_{an} = I_{bn} = I_{cn}$$

$$\text{And } V_{ab} = V_{cb} = V_L$$

$$\text{Hence } P_1 = V_L I_L \cos (30^\circ + \Phi)$$

$$\text{And } P_2 = V_L I_L \cos (30^\circ - \Phi)$$

After simplifying the above

$$\text{So } P_1 + P_2 = 1.732 V_L I_L \cos \Phi$$

$$\text{And } P_2 - P_1 = V_L I_L \sin \Phi$$

$$\tan \Phi = (\sin \Phi / \cos \Phi)$$

$$= 1.732 (P_2 - P_1) / (P_2 + P_1)$$

■Case I ($P_1=P_2$)

When the reading of both wattmeter are equal, the power factor angle is 0° and the power factor is unity, it is a resistive load case.

■Case II ($P_2=0$)

The second wattmeter reading is zero , the power factor angle is 60° and power factor is 0.5 lagging

■Case III (the one wattmeter reading >0 and other wattmeter reading <0)

The case is $90^\circ > \text{power factor angle} > 60^\circ$ and power factor

■Case IV (the sum of both wattmeter reading is zero)

The case is of pure inductive or capacitive load

During observation if any wattmeter shows a negative reading, it is necessary to reverse either its pressure coil or current coil, usually the former. The reading after the reversal of pressure coil is to be taken as negative.

Ex. The input power to a three-phase motor was measured by a two wattmeter method. The readings were 5.2 kW and -1.7kW, and the line voltage was 415V. Calculate:

(a) the total active power,

(b) The power factor,

(c) The line current.

Sol.

(a) Total power = $5.2 - 1.7 = 3.5$ kW

(b) Power factor =

$$\tan \phi = 1.732 \left\{ \frac{5.2 - (-1.7)}{5.2 + (-1.7)} \right\}$$
$$= 3.41$$
$$\phi = 73^{\circ}.39'$$

and power factor = $\cos \phi = 0.281$

(c) Line current

$$3500 = 1.73 * I_L * 415 * 0.281; \quad I_L = 17.3 \text{ A}$$

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